Questions

Q1.

Ultrasound can be used to find cracks in metals.

Figure 13a shows the signals emitted and received when the metal bar has no cracks.

Figure 13b shows the signals emitted and received when the metal bar has a crack.

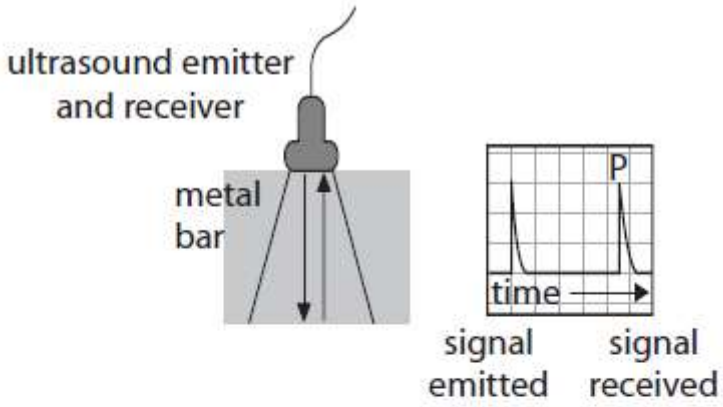

Figure 13a

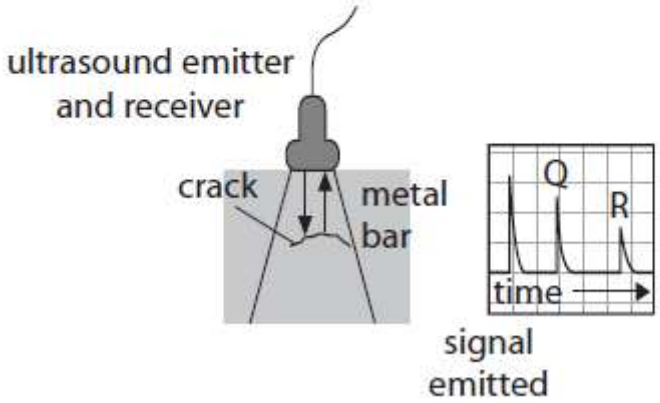

Figure 13b

(i) Explain how the signals in Figure 13a and Figure 13b show that there is a crack in the metal bar in Figure 13b.

(2)

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(ii) Suggest **one** reason why the amplitude of signal R in Figure 13b is smaller than the amplitude of signal P shown in Figure 13a.

(1)

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(Total for question = 3 marks)

Q2.

Explain how sound waves can be used to determine the depth of the ocean directly underneath a boat.

You may draw a diagram to help with your answer.

(6)

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(Total for question = 6 marks)

Q3.

Explain how sound waves can be used to determine the depth of the ocean directly underneath a boat.

You may draw a diagram to help with your answer.

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(Total for question = 6 marks)

Q4.

Explain how vibrations from earthquakes may be used to study the core of the Earth.

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(Total for question = 4 marks)

Q5.

The diagram in Figure 7 shows two students, P and Q, trying to measure the speed of sound in air.

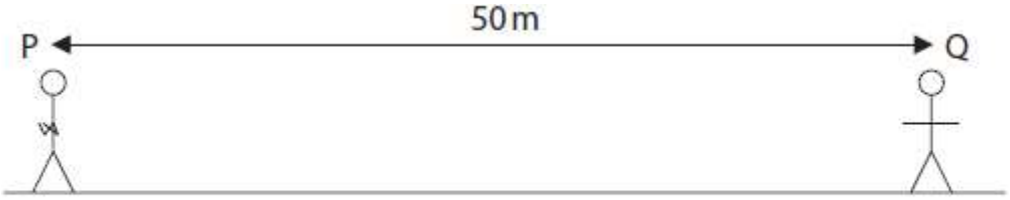

Figure 7

P will clap his hands together.

When Q sees P clap his hands, she will start a timer.

When Q hears the clap, she will stop the timer.

Explain **one** way the students could improve their method.

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(Total for question = 2 marks)

Q6.

Figure 8 shows how a glass prism is used to produce a spectrum from a beam of sunlight.

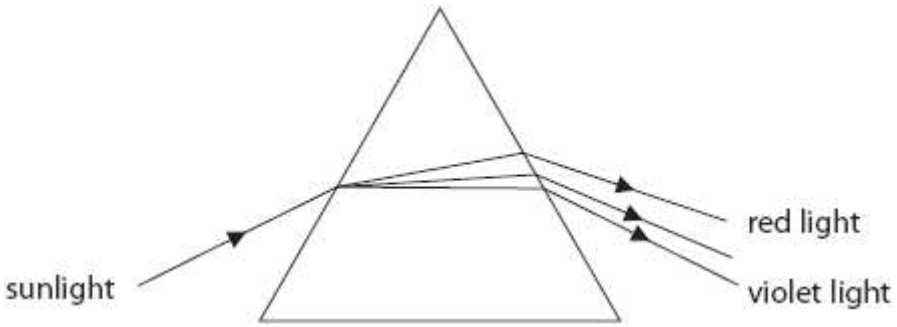

Figure 8

(i) Explain, in terms of refraction and wavelength, the paths taken by the light through the prism.

(3)

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(ii) Describe how a student could develop the procedure shown in Figure 8 to search for evidence that sunlight consists of additional electromagnetic waves with frequencies lower than visible light.

(2)

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(Total for question = 5 marks)

Q7.

The table in Figure 14 gives information about seismic waves, S waves and P waves, produced by an earthquake.

S waves	P waves
transverse	longitudinal
slow moving	fast moving
travel through solids	travel through liquids and solids

Figure 14

Figure 15 shows

- the paths of some S waves and P waves coming from an earthquake
- the types of wave detected in different regions (A, B and C) of the Earth's surface.

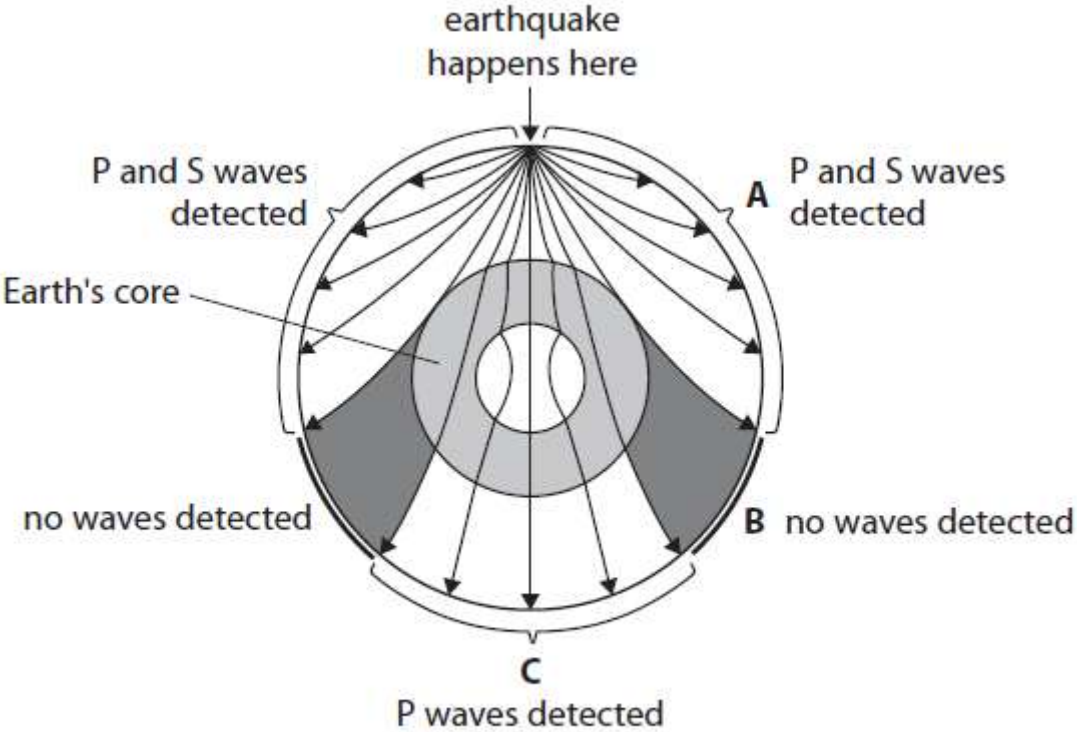

Figure 15

Explain what the diagram in Figure 15 shows about the density of the Earth and the nature of the Earth's core.

Use information from the table in Figure 14 and the diagram in Figure 15 in your answer.

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(Total for question = 6 marks)

Q8.

Figure 6 is a simplified diagram to show radio waves from a transmitter moving upwards, then meeting a boundary between lower and upper layers of the atmosphere.

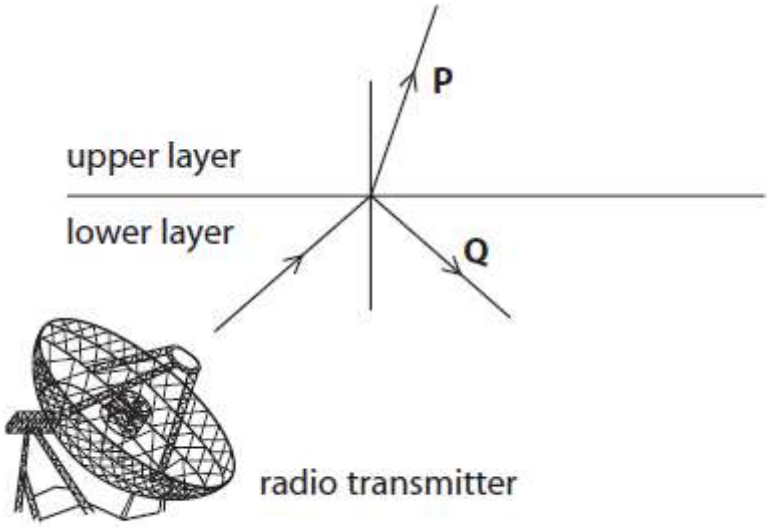

Figure 6

Explain what happens to the radio waves after they meet the boundary between the lower and upper layers as shown in Figure 6.

Your explanation should refer to changes in direction and speed of the waves.

(4)

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(Total for question = 4 marks)

Q9.

This question is about both **infrasound** and **ultrasound**.

Explain why humans cannot hear **infrasound**.

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(Total for question = 2 marks)

Q10.

The ratio of the absorbed energy E_a to incident energy E_i is the coefficient of absorption of sound α.

$$\alpha = \frac{E_a}{E_i}$$

The table in Figure 17 gives the coefficient of absorption for various materials.

material	$\alpha = \frac{E_a}{E_i}$
curtains	0.55
painted walls	0.02
wood floors	0.10
carpeted floors	0.60

Figure 17

Explain why rooms with carpets and curtains are less noisy than rooms without them.

Use the information given in Figure 17 in your answer.

(2)

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(Total for question = 2 marks)

Q11.

The ratio of the absorbed energy E_a to incident energy E_i is the coefficient of absorption of sound α.

$$\alpha = \frac{E_a}{E_i}$$

The table in Figure 17 gives the coefficient of absorption for various materials.

material	$\alpha = \frac{E_a}{E_i}$
curtains	0.55
painted walls	0.02
wood floors	0.10
carpeted floors	0.60

Figure 17

Explain why rooms with carpets and curtains are less noisy than rooms without them.

Use the information given in Figure 17 in your answer.

(2)

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(Total for question = 2 marks)

Mark Scheme

Q1.

	Answer	Additional guidance	Mark
(i)	An explanation linking two of: the crack reflects/bounces back /echoes the signal (1) in a shorter time/ before it reaches the base of the metal (1)	Q is reflection from crack/P and R are reflections from the base	(2) AO3

	Answer	Additional guidance	Mark
(ii)	One from:- less of the pulse reaches the base (1) some energy is absorbed by the crack (1) some energy is reflected by the crack (1)	allow less of the pulse is reflected/bounces off base allow signal/wave/energy/ultrasound for pulse allow signal/wave/pulse/ultrasound for energy allow signal/wave/pulse/ultrasound for energy	(1) AO1

Q2.

Question number	Indicative content	Mark
*	Answers will be credited according to candidate's deployment of knowledge and understanding of the material in relation to the qualities and skills outlined in the generic mark scheme. The indicative content below is not prescriptive and candidates are not required to include all the material which is indicated as relevant. Additional content included in the response must be scientific and relevant. **AO1** • ultrasound pulses used • ultrasound has frequency above 20 kHz • boat has device for producing pulses • ultrasound pulses emitted • ultrasound pulses travel through sea water • pulses reflected by/bounce off/echo off ocean bed • pulses reflected/travel back to boat • time between emission and receipt of pulse measured by device on the boat • velocity of sound in sea water known • use of v=d/t	**6** **AO1.2** A labelled diagram may give some information that is credit worthy
	• speed of sound = distance/time • d=v x t • either distance halved or time halved prior to calculation of distance **accept 'sonar' for ultrasound**	

Level	Mark	Descriptor
	0	• No rewardable material.
Level 1	1-2	• Demonstrates elements of physics understanding, some of which is inaccurate. Understanding of scientific ideas lacks detail. (AO1) • Presents an explanation with some structure and coherence. (AO1)
Level 2	3-4	• Demonstrates physics understanding, which is mostly relevant but may include some inaccuracies. Understanding of scientific ideas is not fully detailed and/or developed. (AO1) • Presents an explanation that has a structure which is mostly clear, coherent and logical. (AO1)
Level 3	5-6	• Demonstrates accurate and relevant physics understanding throughout. Understanding of the scientific ideas is detailed and fully developed. (AO1) • Presents an explanation that has a well-developed structure which is clear, coherent and logical. (AO1)

Level	Mark	Additional Guidance	General additional guidance – the decision within levels e.g. - At each level, as well as content, the scientific coherency of what is stated will help place the answer at the top, or the bottom, of that level.
	0	No rewardable material.	
Level 1	1–2	Additional guidance isolated facts about measuring depth of ocean	Possible candidate responses sound emitted from boat sound goes through the sea sound bounces back / echoes time measured distance=velocity x time uses ultrasound (pulse)
Level 2	3–4	Additional guidance limited description of determination of depth of ocean	Possible candidate responses sound emitted from boat travels to ocean bed and is reflected and time taken from emission to receipt of wave is measured or distance=velocity x time
Level 3	5–6	Additional guidance detailed description of determination of depth of ocean	Possible candidate responses ultrasound pulse emitted from boat ultrasound pulse travels to ocean bed and is reflected time taken for emission to receipt of pulse is measured velocity of sound in sea water known distance travelled = velocity x time halve distance / halve time measured to calculate depth of ocean Any two from three of ultrasound, pulse, halving of distance or time

			required for Level 3.

Q3.

Question number	Indicative content	Mark
*	Answers will be credited according to candidate's deployment of knowledge and understanding of the material in relation to the qualities and skills outlined in the generic mark scheme. The indicative content below is not prescriptive and candidates are not required to include all the material which is indicated as relevant. Additional content included in the response must be scientific and relevant. **AO1** • ultrasound pulses used • ultrasound has frequency above 20 kHz • boat has device for producing pulses • ultrasound pulses emitted • ultrasound pulses travel through sea water • pulses reflected by/bounce off/echo off ocean bed • pulses reflected/travel back to boat • time between emission and receipt of pulse measured by device on the boat • velocity of sound in sea water known • use of v=d/t	**6** **AO1.2** A labelled diagram may give some information that is credit worthy

	• speed of sound = distance/time • d=v x t • either distance halved or time halved prior to calculation of distance **accept 'sonar' for ultrasound**	

Level	Mark	Descriptor
	0	• No rewardable material.
Level 1	1-2	• Demonstrates elements of physics understanding, some of which is inaccurate. Understanding of scientific ideas lacks detail. (AO1) • Presents an explanation with some structure and coherence. (AO1)
Level 2	3-4	• Demonstrates physics understanding, which is mostly relevant but may include some inaccuracies. Understanding of scientific ideas is not fully detailed and/or developed. (AO1) • Presents an explanation that has a structure which is mostly clear, coherent and logical. (AO1)
Level 3	5-6	• Demonstrates accurate and relevant physics understanding throughout. Understanding of the scientific ideas is detailed and fully developed. (AO1) • Presents an explanation that has a well-developed structure which is clear, coherent and logical. (AO1)

Level	Mark	Additional Guidance	General additional guidance – the decision within levels e.g. - At each level, as well as content, the scientific coherency of what is stated will help place the answer at the top, or the bottom, of that level.
	0	No rewardable material.	
Level 1	1–2	Additional guidance isolated facts about measuring depth of ocean	Possible candidate responses sound emitted from boat sound goes through the sea sound bounces back / echoes time measured distance=velocity x time uses ultrasound (pulse)
Level 2	3–4	Additional guidance limited description of determination of depth of ocean	Possible candidate responses sound emitted from boat travels to ocean bed and is reflected and time taken from emission to receipt of wave is measured or distance=velocity x time
Level 3	5–6	Additional guidance detailed description of determination of depth of ocean	Possible candidate responses ultrasound pulse emitted from boat ultrasound pulse travels to ocean bed and is reflected time taken for emission to receipt of pulse is measured velocity of sound in sea water known distance travelled = velocity x time halve distance / halve time measured to calculate depth of ocean Any two from three of ultrasound, pulse, halving of distance or time

			required for Level 3.

Q4.

Question number	Answer	Additional guidance	Mark
	An explanation linking any four from use of seismometers (1) waves can be refracted in the interior of the earth (1) (showing) different densities (1) some seismic waves are longitudinal and some transverse (1) S/transverse waves cannot pass through liquid (1) S wave / transverse wave shadow zone shows part of the earth must be liquid (1) P / longitudinal waves can go through the core/liquid (1) mention S and P waves (1)	S/transverse waves can only pass through solids	**(4) exp**

Q5.

Question Number	Answer	Additional guidance	Mark
	An explanation linking: make the distance between students larger (1) OR viable alternative method such as use microphones / sound sensors / datalogger (to start and stop timer) (1) **with:** to give a more measurable time (1) OR to remove (variable) reaction times (at start and end) / to reduce effect of reaction times / improve accuracy of timing (1)	50 m is too short (a distance to produce a measurable time) gives a longer time – more accurate measurement do not accept 'more accurate' without qualification for either method	**(2)**

Q6.

Question number	Answer	Acceptable	Mark
(i)	An explanation that combines identification - understanding (1 mark) and reasoning/justification - understanding (2 marks): • white light is a mixture of different wavelengths (1) • each wavelength / colour is refracted by a different amount (1) • short wavelengths are refracted more / ORA (1)	ignore colours	(3)

Question number	Answer	Acceptable	Mark
(ii)	An answer that combines the following points to provide a logical description of the method: • Place a thermometer (with blackened bulb) beyond position of red light (1) • Look for rise in temperature (measured by thermometer) (1)		(2)

Q7.

Question number	Indicative content	Mark
	Answers will be credited according to candidate's deployment of knowledge and understanding of the material in relation to the qualities and skills outlined in the generic mark scheme. The indicative content below is not prescriptive and candidates are not required to include all the material which is indicated as relevant. Additional content included in the response must be scientific and relevant. AO1 3 marks AO3 Strand1 3marks When there is an Earthquake **At A** • Longitudinal and transverse waves detected • P arrive before S. • must be travelling in a solid because S waves only travel through solids • the region just under the Earth's crust (mantle) must be a solid • the diagram show the waves refracting so the mantle must be getting more dense as it gets deeper **At B** • No S waves or P waves detected • S waves do not pass through liquid so do not get to B • P waves pass through liquid but are refracted at a solid/ liquid boundary so do not arrive at B • The core of the Earth must be liquid to stop both P and S waves arriving at B **At C** • No S waves arrive at C • P waves arrive at C at different times because the waves have travelled at different speeds. • This suggests the very centre of the Earth would be solid because the P waves that go through the solid would travel faster and arrive before those that just travelled through liquid	**(6)** **AO1** **AO3**

Level	Mark	Descriptor
	0	No rewardable material.
Level 1	1–2	• Demonstrates elements of physics understanding, some of which is inaccurate. Understanding of scientific ideas lacks detail. (AO1) • Interpretation and evaluation of the information attempted but will be limited with a focus on mainly just one variable. Demonstrates limited synthesis of understanding. (AO3)
Level 2	3–4	• Demonstrates physics understanding, which is mostly relevant but may include some inaccuracies. Understanding of scientific ideas is not fully detailed and/or developed. (AO1) • Interpretation and evaluation of the information on both variables, synthesising mostly relevant understanding. (AO3)
Level 3	5–6	• Demonstrates accurate and relevant physics understanding throughout. Understanding of the scientific ideas is detailed and fully developed. (AO1) • Interpretation and evaluation of the information, demonstrating throughout the skills of synthesising relevant understanding. (AO3)

Level	Mark	Additional Guidance	General additional guidance – the decisions within levels
	0	No rewardable material	
Level 1	1-2	Additional guidance Elements of physics present i. e. isolated knowledge of principles, **two** unconnected statements from any section	Possible candidate responses core is liquid/ mantle (crust) is solid waves refract P waves change direction S and P waves travel through mantle(crust) at A, S and P waves detected at B, no waves detected at C, only P waves detected S waves are transverse P waves are longitudinal
Level 2	3-4	Additional guidance Some knowledge of principles with a logical connection made in **ONE** section and a statement in **ONE** other Or From ONE section detailed knowledge of principles with logical connections	Possible candidate response at A, S and P waves are detected so must pass through solid at B no waves are detected, the waves change direction/ Region B is the shadow zone at C, P waves are detected because they go straight through the Earth/can travel through the core in Earth's core there are different densities/waves change direction
Level 3	5-6	Additional guidance Detailed knowledge of principles with logical connections made in **TWO** of the sections	Possible candidate response at A, S and P waves are detected so must pass through solid and refract/curve at B, no waves are detected because S waves do not pass through liquid core and P waves are refracted away by the liquid. At C, only P waves are detected because they can pass through the liquid core and change direction in the core(refracted) In the Earth's core there is refraction of waves/ inner and outer core have different densities

Q8.

Question number	Answer	Additional guidance	Mark
	explanation linking: wave P refracts (towards the normal) (1) because P slows down (1) AND wave Q is reflected (at an equal angle from the boundary) (1) without change of speed of Q (1)	accept 'upper layer' for 'P' accept 'wavelength decreases' accept 'bends' for 'refracts' in this instance accept 'lower layer' for 'Q' accept 'wavelength unchanged' accept 'wave Q bounces off' (at an equal angle) allow one mark for refraction and reflection if no other mark awarded	(4)

Q9.

Question Number	Answer	Additional guidance	Mark
	an explanation linking frequency / Hz (of infrasound) (1) is too low (1) ALTERNATIVE named ear part (1) effect (1)	 below our hearing range below 20 Hz scores 2 marks cochlea / hairs / membrane / ossicles / bones / ear drum not able to vibrate / too stiff / too flexible / too thick / too thin	**(2)** **AO2.1**

Q10.

Question	Answer	Additional guidance	Mark
	An explanation linking more absorption giving less reflection of sound (coefficient of) absorption / α is greater (for carpets / curtains than walls / wood floors) or more of the incident energy/sound is absorbed (1)	ignore references to insulation	**2** **AO3.1**

	(therefore) less of the incident energy /sound is reflected (back into the room) or wooden floors and painted walls reflect sound (1)	accept echo/bounces around (the room) for reflection	

Q11.

Question	Answer	Additional guidance	Mark
	An explanation linking more absorption giving less reflection of sound (coefficient of) absorption / α is greater (for carpets / curtains than walls / wood floors) or more of the incident energy/sound is absorbed (1)	ignore references to insulation	**2** **AO3.1**

	(therefore) less of the incident energy /sound is reflected (back into the room) or wooden floors and painted walls reflect sound (1)	accept echo/bounces around (the room) for reflection	